A conditional refinement of the Subset Sum exponent to 0.487676

Joshua Wharton

QFEX

joshua@qfex.com

20 October 2026

Abstract

We give a parameter and certificate refinement of the claimed $O(2^{0.4877n})$ Subset Sum construction in the pinned manuscript $\mathcal{B}$ [1]. The internal work exponent is reduced from 0.48769 to 0.48767595. A transfer argument preserves the existing 43 filters and derives the required new moment bounds from their 430 original certificates, without new quadrature. Together with exact rational checks of the alias constraints, this gives the conditional bound $O_c(2^{0.487676n})$ for every fixed polynomial input-bit-length bound $b \le n^c$. The qualification is essential: the algorithmic lemmas and original moment inequalities of $\mathcal{B}$ are inherited premises, not independently verified here. This note is not a formal proof-assistant verification, a literature-priority claim, or a practical speedup claim.

1 Statement and dependencies

Let $\mathcal{B}$ denote `subset-sum/PROOF_0.4877.tex` in `a-kapila/maths-proofs`, pinned at commit `44c34961b56784c220a6e913cc068d57dc37c5a3` [1]. That manuscript relies on the 4 October 2026 preprint *Subset Sum in Time* $O(2^{0.49n})$ [2]. The intended specification is the Subset Sum benchmark [3]; no assertion about its current leaderboard is needed for this note.

The inherited premises are the isolation and implementation lemmas, the preliminary-search and alias arguments under their numerical side conditions, the parameterised checksum construction, finite-circle convergence, and the certified continuous moment bounds of $\mathcal{B}$. References to numbered statements of $\mathcal{B}$ refer to that pinned version.

Theorem 1 (Conditional refinement)**.** *Assume these inherited premises of $\mathcal{B}$ are valid. There is a single uniform randomised classical word-RAM program which halts on every input and is correct with probability at least $2/3$. For each fixed positive integer c, there are constants $C_c, N_c > 0$ such that, for every input with $n \ge N_c$ and $b \le n^c$, every random execution uses at most*

$$C_c\, 2^{0.487676n}$$

word operations. Here b is the largest bit length of an input integer or the target, with the conventions of $\mathcal{B}$, and the word length is

$$w_{\mathrm{RAM}} = \lceil 4(n + b + \log_2(n+2)) \rceil .$$

The program does not depend on c.

The baseline internal constants are

$$\tau_b = 0.48769, \qquad \eta_b = 0.51613, \qquad \Delta_b = 0.02844, \qquad \Gamma_b = 0.45925. \tag{1}$$

Its displayed exponent 0.4877 absorbs overhead beyond the linear exponent 0.4876902. The refinement below changes the internal work exponent; it does not merely spend the baseline's rounding slack.

2 Parameters and input guard

All terminating decimals in the parameter table are exact rational numbers. The machine-readable values are in `parameters.json`.

Parameter	Exact value	Meaning
τ	0.48767595	Internal work exponent
η	0.51612858	Large-prime exponent
a	0.008145205148	Mask fraction
μ	0.475442899283	Large-block fraction
γ	0.060035950245	Small-prime parameter
l	0.483780402235	Left-block fraction
$r = 1 - \mu - l$	0.040776698482	Right-block fraction
ε	10^{-11}	Pattern accuracy
ζ	10^{-11}	Preliminary-search slack

The new checksum budgets are

$$\Delta = \eta - \tau = 0.02845263, \qquad \Gamma = 2\tau - \eta = 0.45922332. \tag{2}$$

Retain the baseline's 43 filters, filter-selection rule, isolation transformation, and final decision rule. Construct checksum tables and choose the sampling fraction using the *new* budgets. The regions used in the certificate analysis will retain the *old* boundaries.

Replace the input guard by

$$10^{11} \lceil \log_2(b+1) \rceil \le n. \tag{3}$$

When the guard fails, or n is below a sufficiently large fixed cutoff, use the exact meet-in-the-middle fallback of $\mathcal{B}$. For each fixed c, the guard eventually holds whenever $b \le n^c$. On inputs passing it,

$$b + 2 \le 2^{10^{-11} n + 1}. \tag{4}$$

The bit-length factor in the alias analysis therefore consumes at most 10^{-11} of the linear-exponent budget.

3 Alias work and coverage

Write

$$h(x) = -x \log_2 x - (1-x) \log_2 (1-x).$$

The strict alias constraints of $\mathcal{B}$, Section 2.2, become

$$\gamma < \tfrac{1}{2} h(2a), \tag{C1}$$

$$l + \big(h(a) - \gamma\big)\mu < \tau, \tag{C2}$$
$$r + (1 - \gamma)\mu < \tau, \tag{C3}$$
$$1 - \mu + \big(h(a) + h(\tfrac{1}{2} - a) - \gamma\big)\mu - \eta < \tau, \tag{C4}$$
$$h(\tfrac{1}{2} - a)\mu + \tfrac{1}{2} - \tau < \tau, \tag{C5}$$
$$0 < r < \tau, \qquad l < \tau, \tag{C6}$$
$$\mu(1 - 2a) < l, \tag{C7}$$
$$\tfrac{\mu}{2} \log_2 3 < \eta. \tag{C8}$$

These respectively control representation coverage, left lists, right value classes, matching, mask enumeration, value-map construction, deficiency accounting, and reduction on half-blocks.

The accompanying exact-rational interval checker verifies all eight constraints. The tightest five have the following approximate slacks:

Constraint	Positive slack
Coverage, (C1)	$1.84806989556 \times 10^{-8}$
Left list, (C2)	$1.84818109678 \times 10^{-8}$
Right value classes, (C3)	$1.84806927342 \times 10^{-8}$
Matched pairs, (C4)	$1.84813377432 \times 10^{-8}$
Right-mask enumeration, (C5)	$1.84815267754 \times 10^{-8}$

Every tested alias and auxiliary size inequality has slack at least

$$\delta_{\text{alias}} = 1.8480692734174335 \times 10^{-8}. \tag{5}$$

The auxiliary checks include

$$0 < a < \tfrac{1}{4}, \qquad \mu(1 - 2a) < \tfrac{1}{2}, \qquad \tfrac{1}{2}h(2a) + h(a) < 1 - 2a, \tag{6}$$

and $h(a) < h(\frac{1}{2} - a)$. The layer inequality supplies the combinatorial size condition in the averaged small-layer argument of $\mathcal{B}$.

In particular,

$$\zeta + 10^{-11} = 2 \times 10^{-11} < \delta_{\text{alias}}.$$

Thus the preliminary-search slack and input-bit-length factor fit strictly inside the alias work margin. Under the inherited alias premises, the preliminary-search, coverage, expected-work and cap arguments carry over with the new constants. The capped alias cost is

$$2^{\tau n + O(n^{0.7})} \operatorname{poly}(n + b). \tag{7}$$

Strict inequalities absorb the integer rounding of block and mask sizes for sufficiently large n, as in $\mathcal{B}$.

4 Transferring the checksum certificates

Optimising the alias constraints alone would not establish the claimed refinement. The checksum budgets change as well, so both moment bounds and their domination of sampled patterns must be preserved.

Use the baseline score functions $u_v(x), w_v(x)$ for $v \in \{0, \dots, 42\}$. For each pattern, the selected filter maximises its approximate score U_v; write U, W for that filter's scores. The baseline guarantees $W \geq 0$. The new sampling fraction is

$$\alpha = \begin{cases} 1, & W \leq \Gamma, \\ \Gamma / W, & W > \Gamma. \end{cases} \tag{8}$$

A sampled pattern satisfies $\alpha U < \Delta$.

Set

$$q = \frac{\Gamma}{\Gamma_b}, \qquad A_b = \frac{\Delta_b}{\Gamma_b}, \qquad A = \frac{\Delta}{\Gamma}, \qquad K = \frac{\Delta}{q} - \Delta_b. \tag{9}$$

These constants satisfy

$$0 < q < 1, \qquad K > 0, \qquad A - A_b = K/\Gamma_b, \tag{10}$$

with approximate values

$$\begin{aligned} q &= 0.99994190528034839412 0849\ldots, \\ K &= 0.00001428304359630517021652994\ldots, \\ A - A_b &= 0.00003110080260491055028095796\ldots. \end{aligned}$$

Retain the old region boundaries Γ_b times

$$1,\ 1.25,\ 1.6,\ 2,\ 2.6,\ 3.5,\ 5,\ 8,\ 15,\ \infty.$$

These are boundaries for the proof, not for constructing the new tables.

Each old certificate has a nonnegative multiplier vector z and an affine row vector $G_b(x)$. Write

$$I_b = \int_{\mathbb{T}} M(x)^2\, 2^{z \cdot G_b(x)}\, dx, \qquad M(x)^2 = 2 + 2\cos x,$$

where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ has normalised measure. Comparison rows $u_v - u_j$ correspond to other filters j and are nonnegative at the pattern level because v was selected to maximise U_v.

Lemma 2 (Certificate transfer)**.** *Assume* (10)*. The old certificates can be modified as follows. Each modification retains the required domination for sampled patterns in the old region.*

1. *For $W \le \Gamma_b$, replace the first row $\Delta_b - u$ by $\Delta/q - u$, leaving all multipliers unchanged. Then $I' = 2^{z_0 K} I_b$.*
2. *For $\Gamma_b L < W \le \Gamma_b H$ with finite H and $L \ge 1$, replace $A_b w - u$ by $Aw - u$ and replace the multiplier of $\Gamma_b H - w$ by $z_2 + z_0(A - A_b)$. Then $I' = 2^{z_0 K H} I_b$.*
3. *For $W > 15\Gamma_b$, replace $A_b w - u$ by $Aw - u$ and leave all multipliers unchanged. If $w_v(x) \le B$ on the real circle, then $I' \le 2^{B z_0 (A - A_b)} I_b$.*

Here I' denotes the integral of the modified certificate.

Proof. In the low region, (8) gives $q \le \alpha \le 1$. Put $\Theta = \Delta/q = \Delta_b + K$. Sampling implies $U < \Delta/\alpha \le \Theta$. The old condition $z_0 \ge 1$ gives

$$-\alpha U = -\Delta + \alpha(\Delta/\alpha - U) \le -\Delta + z_0(\Theta - U). \tag{11}$$

All remaining region and comparison rows are nonnegative at the pattern level. Pointwise, the modified exponent is the old exponent plus $z_0 K$, proving the first integral identity. This includes the interval $\Gamma < W \le \Gamma_b$, where α is less than one.

In a bounded region the old first three rows are

$$(A_b w - u,\ w - \Gamma_b L,\ \Gamma_b H - w).$$

The multiplier increase is nonnegative. The change in the total exponent is exactly

$$z_0(A - A_b)w + z_0(A - A_b)(\Gamma_b H - w) = z_0 K H. \tag{12}$$

The old condition $z_0 L \geq 1$ implies $\alpha < q/L < 1/L \leq z_0$. Moreover $\alpha AW = \Delta$ and sampling implies $AW - U > 0$. Hence

$$-\alpha U = -\Delta + \alpha(AW - U) \leq -\Delta + z_0(AW - U). \tag{13}$$

The remaining rows are again nonnegative. This proves the second case.

In the unbounded region, $15z_0 \geq 1$ and $\alpha < q/15$ give the same domination argument. The exponent changes by $z_0(A - A_b)w(x)$, which is at most $Bz_0(A - A_b)$. Positivity of $M(x)^2$ on the real circle then proves the final integral bound. □

The lemma establishes ideal pattern-level domination. The effect of approximating the coordinate scores is accounted for explicitly in Section 5; it is not silently omitted from the transfer.

4.1 A pointwise bound for the tail

Lemma 3 (Coarse score bound)**.** *For each of the 43 baseline filters,* $|w_v(x)| \leq 256$ *on the real circle.*

Proof. The finite data in Appendix A of $\mathcal{B}$ give eight cosine coefficients for each of r_v, s_v, each of absolute value at most 2, and a power parameter $1 \leq \kappa_v \leq 2$. Suppress the filter index. Thus $|r(x)|, |s(x)| \leq 16$. With normalised integrals, put

$$F = 2^r, \qquad G = 2^s, \qquad Z_f = \int F, \qquad Z_g = \int G,$$
$$Z_{f,\kappa} = \int F^\kappa, \qquad Z_{g,\kappa} = \int G^\kappa.$$

Let

$$\mathcal{P}(x) = \int F(x+y)G(y)\,dy,$$
$$\mathcal{Y}(x) = \int F(x+y)G(y)(r(x+y) + s(y))\,dy.$$

The identities inherited from $\mathcal{B}$ are

$$\begin{aligned} L(x) &= \log_2 \mathcal{P}(x) - \log_2(Z_f Z_g), \\ B_*(x) &= \mathcal{Y}(x)/\mathcal{P}(x) - \log_2(Z_f Z_g), \\ H_* &= \log_2 Z_{f,\kappa} + \log_2 Z_{g,\kappa} - \kappa \log_2(Z_f Z_g), \\ w(x) &= (2-\kappa)B_*(x) - 2L(x) + H_*. \end{aligned}$$

Positivity gives $|\log_2(Z_f Z_g)| \leq 32$ and $|\log_2 \mathcal{P}(x)| \leq 32$. Also $\mathcal{Y}/\mathcal{P}$ is a weighted average of $r(x+y) + s(y)$, so its absolute value is at most 32. Consequently

$$|L| \leq 64, \qquad |B_*| \leq 64, \qquad |H_*| \leq 64\kappa.$$

Because $2 - \kappa \geq 0$,

$$|w| \leq (2-\kappa)64 + 128 + 64\kappa = 256.$$

□

5 Moment margins and approximation errors

The following rounded-down moment margins are inherited from Proposition B and Section 8.2 of $\mathcal{B}$. They are assumptions of this note, not quadratures evaluated by the new checker.

Old region	Lower bound on $\eta_b - \log_2 I_b$
ρ_0: $W \le \Gamma_b$	0.00025
ρ_1: $\Gamma_b < W \le 1.25\Gamma_b$	0.00025
$\rho_2, \dots, \rho_8$	4
ρ_9: $W > 15\Gamma_b$	17

The finite source data give

$$\max_v z_0^{(v,\rho_0)} = 17.211, \qquad \max_v z_0^{(v,\rho_1)} = 8.4537.$$

The baseline bounds every multiplier-vector sum by $Z = 418.36$. These values and the transcribed filter data are recorded in `source_facts.json`. The optional source audit checks the transcriptions against a supplied local copy of the pinned TeX source.

Proposition 4 (Transferred continuous moments)**.** *Assuming the old moment margins above, every modified certificate has $\eta - \log_2 I' > 10^{-6}$. Every modified multiplier sum is less than* 419.

Proof. Since $\eta - \eta_b = -0.00000142$, Lemmas 2 and 3 give lower bounds

$$\begin{aligned}
m_0 &= 0.00025 + (\eta - \eta_b) - 17.211K > 0.0000027545366639917,\\
m_1 &= 0.00025 + (\eta - \eta_b) - 8.4537(1.25)K > 0.0000976492929373937,\\
m_{2:8} &= 4 + (\eta - \eta_b) - 418.36(15)K > 3.9103667682,\\
m_9 &= 17 + (\eta - \eta_b) - 418.36(256)(A - A_b) > 13.6690976448.
\end{aligned}$$

All are strictly greater than 10^{-6}. For the bounded regions use $z_0 \le Z$ and $H \le 15$; for the tail use $z_0 \le Z$. The only multiplier increase is $z_0(A - A_b)$, so every modified sum is at most $Z(1 + A - A_b) < 419$. These are exact rational inequalities verified by `verify.py`. □

There are only finitely many new, fixed affine certificates. The finite-circle convergence premise of $\mathcal{B}$ therefore supplies a finite-circle moment margin of 10^{-6} for all sufficiently large primes. The strict continuous margin in Proposition 4 is what permits this invocation of convergence; convergence itself is not proved by the scalar checker.

The coordinate-average errors inherited from the baseline are $|U_v - \bar{u}_v| \le 3\varepsilon$ and $|W_v - \bar{w}_v| \le 4\varepsilon$. Each modified affine row has error at most 6ε. In particular, $Aw - u$ has error at most $(4A + 3)\varepsilon < 4\varepsilon$ since $A < 1/4$; comparison rows have error at most 6ε. Fixed region constants introduce no additional error.

The domination proof and baseline mean-square-error argument therefore leave the positive exponent

$$10^{-6} - (18 + 6 \cdot 419)\varepsilon = 9.7468 \times 10^{-7}. \tag{14}$$

For each sampled pattern the corresponding mean-square error, restricted to the baseline good-table event, is at most

$$2^{-(9.7468\times 10^{-7})n + O(1)}.$$

The $2^{O(\sqrt{n})}$ patterns and polynomially many targets are absorbed by this strict exponential decay. The inherited good-table probability, Cauchy–Schwarz, Markov's inequality and a union bound give the same constant absolute-error guarantees as $\mathcal{B}$, with probability $1 - o(1)$.

6 Correctness and pure exponential time

Proof of Theorem 1. Use the baseline program with the constants and guard of Section 2, retaining its bounded samplers, caps and exact fallback. The preliminary search and alias procedure remain valid under the inherited premises and the strict checks in Section 3. Sections 4–5 supply the replacement checksum moment and error analysis.

The exact modular checksum minus the exact alias phase sum is the phase sum over subsets representing the unreduced target. On a NO instance it is zero. When isolation produces a unique representing subset for some target, its absolute value is one. The threshold $1/2$ therefore separates these cases on the inherited success events. Isolation and the vanishing failure probabilities give correctness at least $2/3$ above a sufficiently large fixed cutoff. The fallback is exact below the cutoff and whenever the guard fails.

The work caps and bounded samplers ensure halting and the resource bound on every random outcome, not merely in expectation. Prime generation costs $2^{\eta n/2}\,\mathrm{poly}(n+b)$, which is below the work budget. The total cost has the inherited form

$$T(n,b) \le 2^{(\tau+20\varepsilon)n+O(n^{0.7})}\,\mathrm{poly}(n+b) = 2^{0.4876759502n+O(n^{0.7})}\,\mathrm{poly}(n+b). \tag{15}$$

For fixed c and $b \le n^c$, the guard eventually holds and

$$0.487676 - (\tau + 20\varepsilon) = 4.98 \times 10^{-8} > 0.$$

This linear-exponent gap absorbs both the subexponential factor and $\mathrm{poly}(n+n^c)$. Thus (15) is $O_c(2^{0.487676n})$. The inherited address-space argument applies with the same word length. All cutoffs and program parameters are fixed independently of c. □

Remark. The cutoff and hidden constants may be very large. This argument gives an asymptotic conditional refinement, not a demonstrated practical speedup. It does not establish that 0.487676 is optimal even within the inherited framework.

7 Reproducibility and verification scope

The accompanying files are `paper.tex`, `verify.py`, `parameters.json`, `source_facts.json`, and `verification.txt`. A `Makefile` provides the convenience targets `verify` and `pdf`; neither target accesses the network. To repeat the scalar checks, run

```
python3 verify.py
```

An optional audit of the finite baseline data is available as

```
python3 verify.py --baseline /path/to/PROOF_0.4877.tex
```

The source file should be obtained from the exact commit in reference [1]. This optional audit checks transcribed data and multiplier rows; it does not check the original integrals or algorithmic correctness, nor authenticate the downloaded file's provenance.

The checker uses only Python's standard library. Every inequality decision uses `fractions.Fraction`. Logarithms are enclosed by range reduction and, for $1 \le y \le 2$, the expansion

$$\ln y = 2\sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} + R_m, \qquad z = \frac{y-1}{y+1}, \tag{16}$$

with the exact rational enclosure

$$0 \le R_m \le \frac{2z^{2m+1}}{(2m+1)(1-z^2)}. \tag{17}$$

The implementation uses $m = 40$. Division by an interval for $\ln 2$ gives intervals for $\log_2$, and interval operations enclose the entropy values. Displayed decimal values are rounded outwards and are not used to decide whether inequalities hold.

The included run was repeated on 20 October 2026 and passed the new alias constraints, auxiliary size conditions, guard margin, moment-transfer bounds, multiplier-sum budget, error exponent and pure-exponent gap. It uses the supplied finite-data transcriptions. The optional full-source audit and the original 430 quadratures were *not* run as part of this verification. All original algorithmic and analytic premises remain explicit dependencies.

Provenance and status. This research note and its accompanying checker were prepared with ChatGPT. The new certificate-transfer argument is a written derivation; the script checks its scalar side conditions. Neither the script nor this note is a Lean verification or an independent verification of the underlying Subset Sum algorithm. No global optimality or literature priority is claimed.

References

[1] `a-kapila/maths-proofs`, *Subset Sum in time $O(2^{0.4877n})$: a complete proof*, 12 October 2026. File `subset-sum/PROOF_0.4877.tex`, pinned commit `44c34961b56784c220a6e913cc068d57dc37c5a3`. `https://github.com/a-kapila/maths-proofs/blob/44c34961b56784c220a6e913cc068d57dc37c5a3/subset-sum/PROOF_0.4877.tex`. The title is the source's title; this note treats its results as inherited premises.

[2] OpenAI, *Subset Sum in Time $O(2^{0.49n})$*, OpenAI Math Release preprint, 4 October 2026. Repository entry pinned at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. `https://github.com/openai/math/blob/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/Subset-Sum-in-Time-2-power-0-49n-October-4-2026/README.md`.

[3] Big O Speedrun, *Subset Sum benchmark*. `https://bigospeedrun.com/subset-sum`.